

Geometry
Unit 7
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

$\triangle PGL$ is a right triangle where $m\angle LPG = 90^\circ$, $LP = 8$, and $LG = 17$.

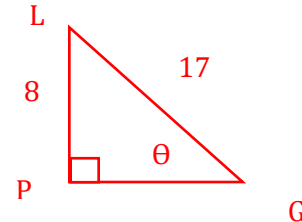
1. Determine $m\angle LGP$ using a sine ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\sin \theta = \frac{8}{17}$$

$$\theta = \sin^{-1}\left(\frac{8}{17}\right)$$

$$\theta = 28.072^\circ$$

$$m\angle LGP = 28.072^\circ$$



$\triangle BLC$ is a right triangle where $m\angle BCL = 90^\circ$, $BC = 33$, and $LC = 56$.

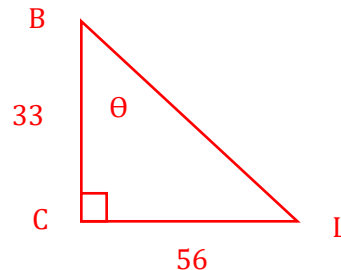
2. Determine $m\angle CBL$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan \theta = \frac{56}{33}$$

$$\theta = \tan^{-1}\left(\frac{56}{33}\right)$$

$$\theta \approx 59.490^\circ$$

$$m\angle CBL \approx 59.490^\circ$$



$\triangle MDZ$ is a right triangle where $m\angle DZM = 90^\circ$, $MD = 41$, and $DZ = 9$.

3. Determine the length of $\overline{ZM}$. If necessary, round your answer to the nearest thousandth.

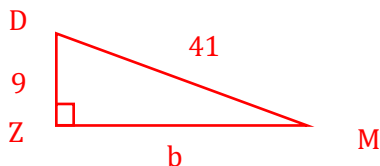
$$9^2 + b^2 = 41^2$$

$$b = \sqrt{1600}$$

$$ZM = 40$$

$$81 + b^2 = 1681$$

$$b = 40$$



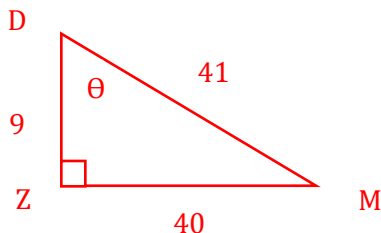
4. Determine $m\angle MDZ$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan \theta = \frac{40}{9}$$

$$\theta = \tan^{-1}\left(\frac{40}{9}\right)$$

$$\theta = 77.320^\circ$$

$$m\angle MDZ \approx 77.320^\circ$$



$\triangle TKB$ is a right triangle where $m\angle KTB = 90^\circ$, $KT = 92$, and $KB = 115$.

5. Determine the length of $\overline{TB}$. If necessary, round your answer to the nearest thousandth.

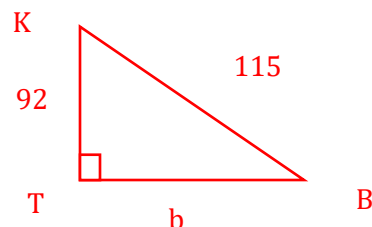
$$92^2 + b^2 = 115^2$$

$$8464 + b^2 = 13,225$$

$$b = \sqrt{4761}$$

$$b = 69$$

$$TB = 69$$

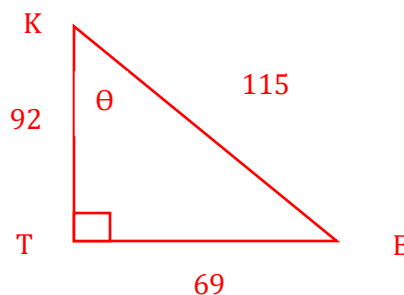


6. Determine $m\angle TKB$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan \theta = \frac{69}{92}$$

$$\theta = \tan^{-1} \left(\frac{69}{92} \right) \quad m\angle TKB \approx 36.870^\circ$$

$$\theta \approx 36.870^\circ$$

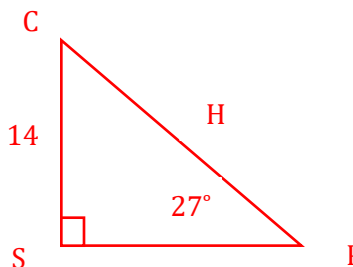


$\triangle SPC$ is a right triangle where $m\angle CSP = 90^\circ$, $m\angle CPS = 27^\circ$, and $CS = 14$.

7. Determine the length of $\overline{CP}$ using a sine ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\sin 27^\circ = \frac{14}{H}$$

$$H = \frac{14}{\sin 27^\circ} \approx 30.838 \quad CP \approx 30.838$$

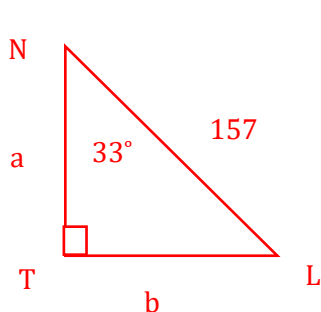


8. Explain why the tangent ratio cannot be used to determine the length of $\overline{CP}$.

Tangent is opposite over adjacent and does not include the hypotenuse. It could be used to find $\overline{SP}$ and then the Pythagorean Theorem could be used to find $\overline{CP}$, but the tangent alone cannot be used to find $\overline{CP}$ directly.

$\triangle NLT$ is a right triangle where $m\angle NTL = 90^\circ$, $m\angle TNL = 33^\circ$, and $NL = 157$.

9. Determine the length of $\overline{TL}$. Show your work. If necessary, round your answer to the nearest thousandth.



$$\sin 33^\circ = \frac{b}{157}$$

$$b = 157 \sin 33^\circ \quad TL \approx 85.508$$

$$\approx 85.508$$

10. Determine the length of $\overline{NT}$. Show your work. If necessary, round your answer to the nearest thousandth.

$$\cos 33^\circ = \frac{a}{157}$$

$$a = 157 \cos 33^\circ \quad NT \approx 131.671$$

$$\approx 131.671$$